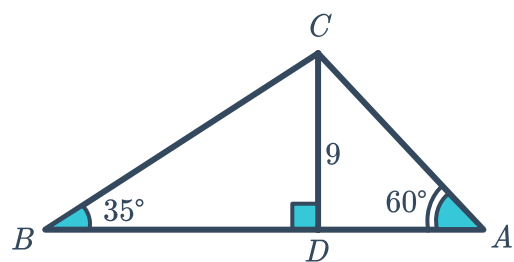


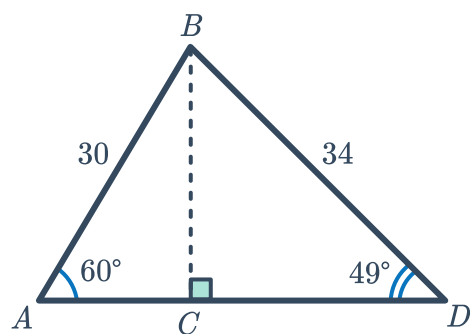
Unknown side lengths



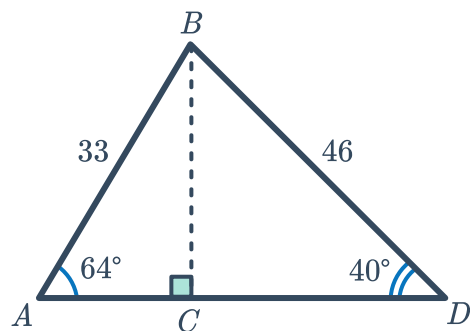
- 1 Consider the diagram shown:
Find the length of AB , correct to two decimal places.



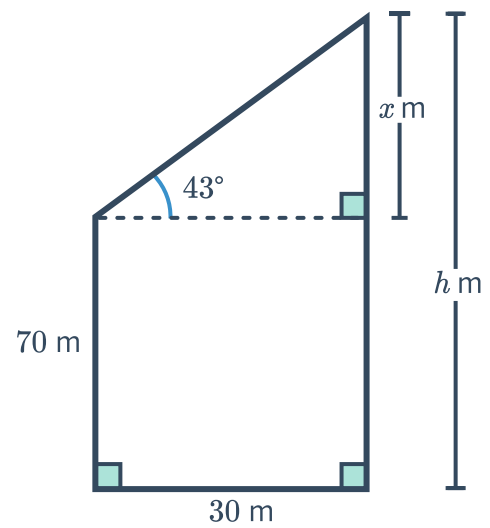
- 2 Consider the following diagram:
Find the length of AD to two decimal places.



- 3 Consider the following diagram:
Find the length of AD to two decimal places.



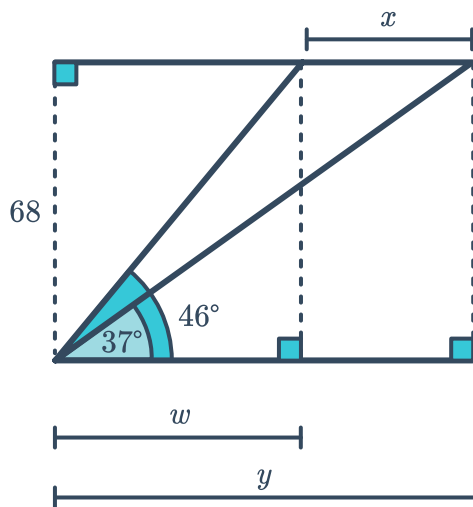
- 4 Consider the following diagram:
Find the values of x and h to the nearest whole number.



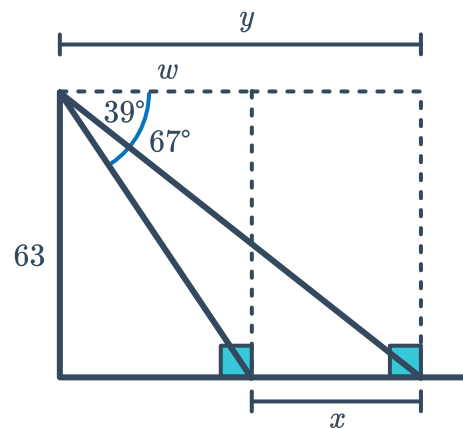
- 5 For each of the following diagrams.:

- Find y , correct to two decimal places.
- Find w , correct to two decimal places.
- Hence, find the value of x , correct to one decimal place.

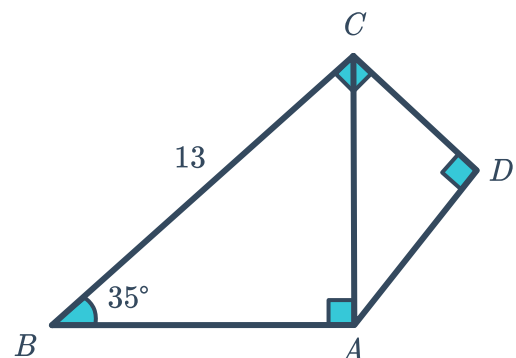
a



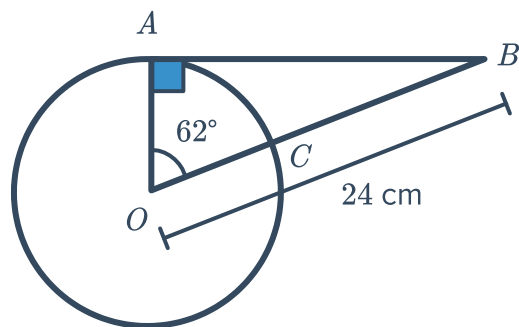
b



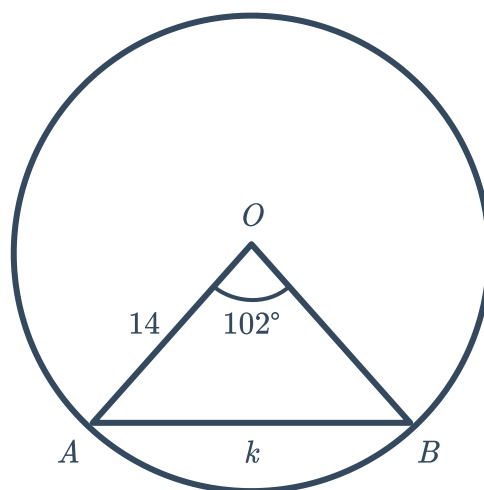
- 6 Consider the given diagram:
Find the length of CD , to one decimal place.



- 7 AB is a tangent to a circle with centre O . OB is 24 cm long and cuts the circle at C . Find the length of BC to the nearest centimetre.

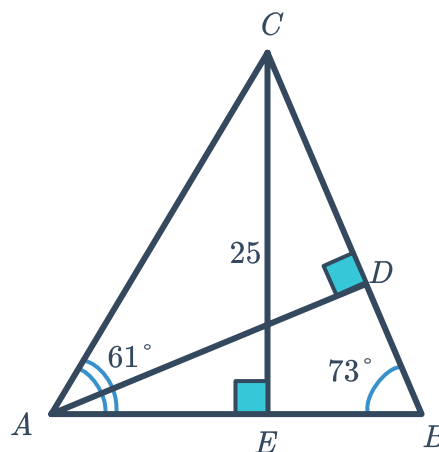


- 8 Consider the following diagram:
Find the value of k . Round your answer to two decimal places.



- 9 In the following diagram, $\angle CAE = 61^\circ$, $\angle CBE = 73^\circ$ and $CE = 25$.

- a Hence, find the length of AB , correct to two decimal places.
b Find the length of BD , correct to one decimal place.

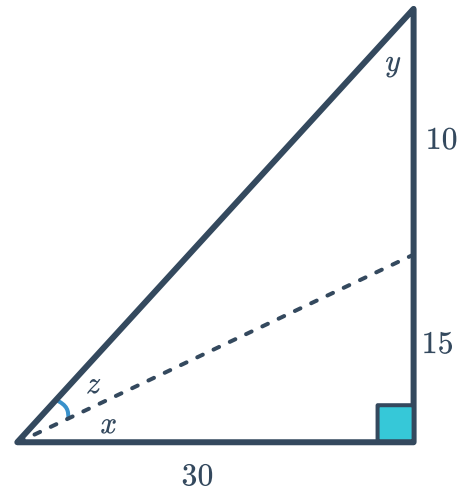




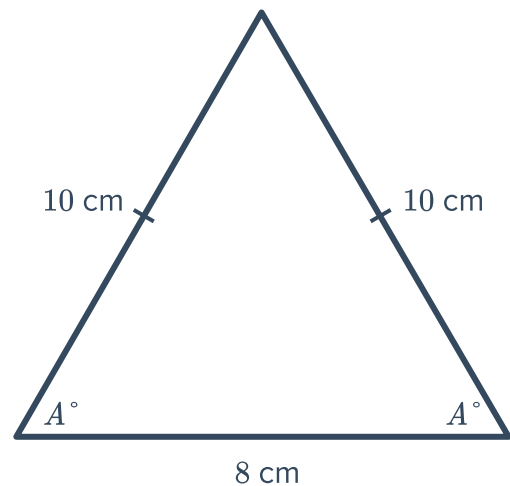
Unknown angles

10 Consider the given figure:

- a Find x , correct to two decimal places.
- b Find y , correct to two decimal places.
- c Find z , correct to two decimal places.

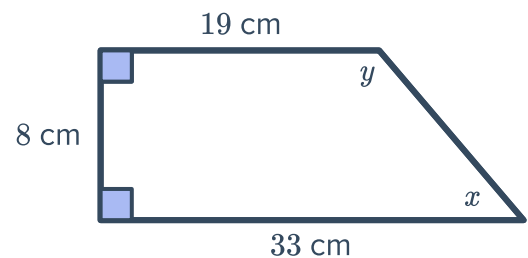


11 An isosceles triangle has equal side lengths of 10 cm and a base of 8 cm as shown. Calculate the size of angle A to one decimal place.



12 Consider the following figure:

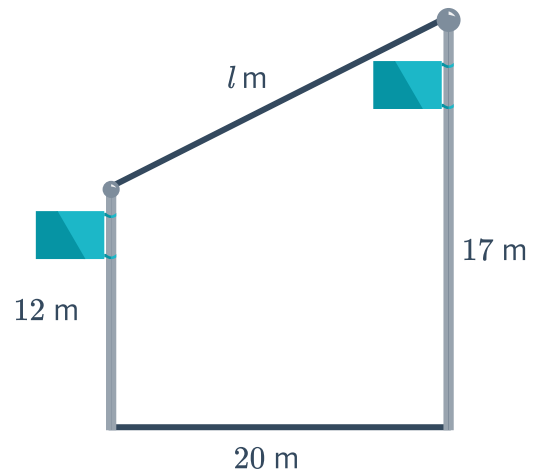
- a Find the size of angle x in degrees, correct to two decimal places.
- b Find the size of angle y in degrees, correct to two decimal places.



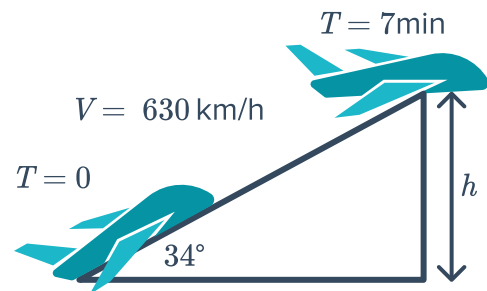


Applications

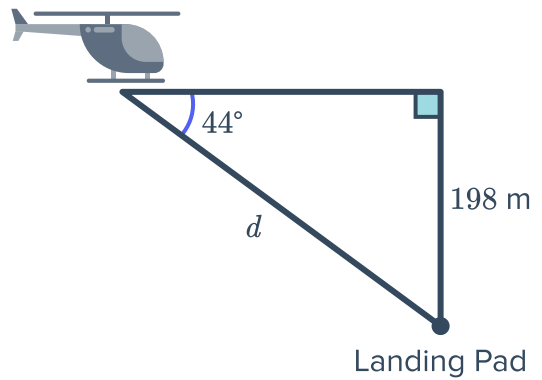
- 13** Two flag posts of height 12 m and 17 m are erected 20 m apart. Find the length, l , of the string needed to join the tops of the two posts. Round your answer to one decimal place.



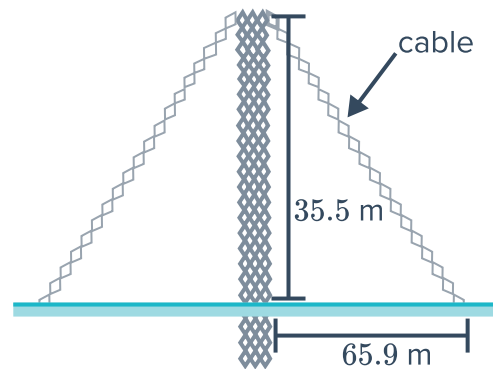
- 14** A jet takes off and leaves the runway at an angle of 34° . It continues to fly in this direction for 7 min at a speed of 630 km/h before levelling out.
- a** Find the distance in metres covered by the jet just before levelling out.
 - b** If the height of the jet just before levelling out is h m, calculate h . Round your answer to the nearest metre.



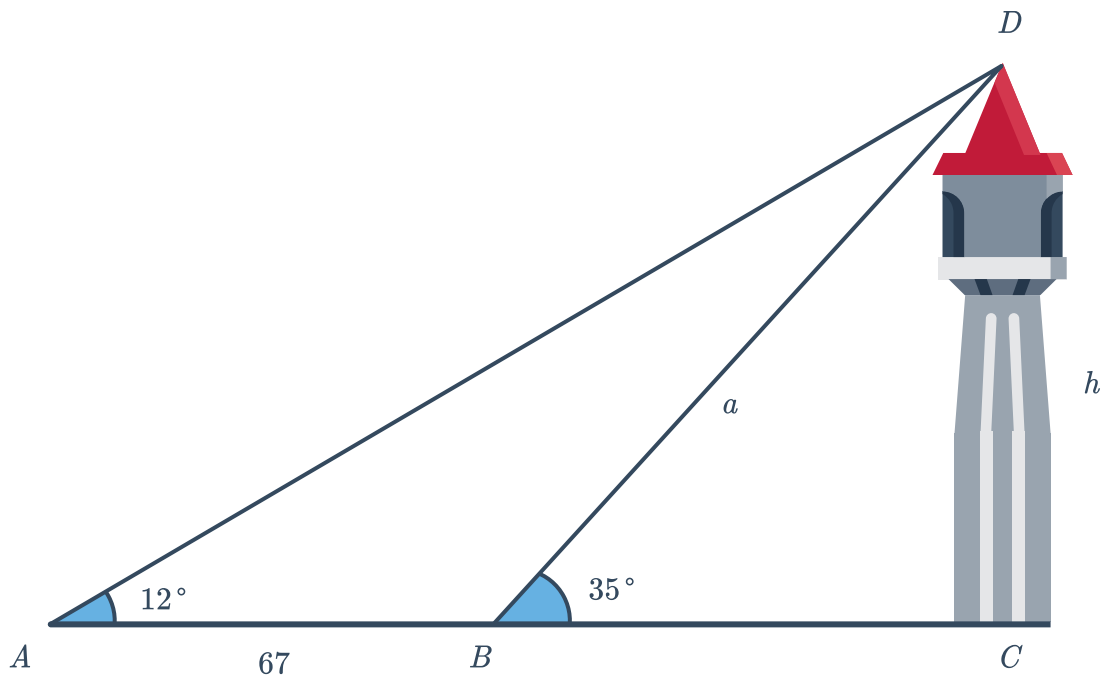
- 15 A helicopter is flying at an altitude of 198 m. Its landing pad is at an angle of depression of 44° . Determine the distance, d , between the helicopter and the landing pad. Round your answer to the nearest whole number.



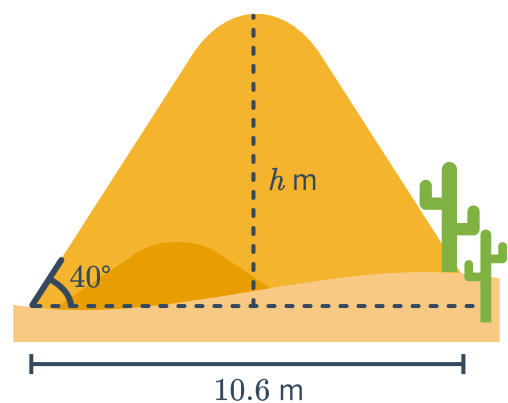
- 16 A suspension bridge is being built. The top of the concrete tower is 35.5 m above the bridge and the connection point for the main cable is 65.9 m from the tower. Assume that the concrete tower and the bridge are perpendicular to each other.
- Find the length of the cable to two decimal places.
 - Find the angle the cable makes with the road to two decimal places.



- 17 Mae observes a tower at an angle of elevation of 12° . The tower is perpendicular to the ground. Walking 67 m towards the tower, she finds that the angle of elevation increases to 35° as shown in the diagram below:



- a Calculate the size of $\angle ADB$.
- b Find the length of the side a . Round your answer to two decimal places.
- c Find the height, h , of the tower. Round your answer to one decimal place.
- 18 A sand pile has an angle of 40° and is 10.6 m wide. Find the height of the sand pile, h , to one decimal place.



- 19 A girl is flying a kite that is attached to the end of a 23.4 m length of string. The angle between the string and the vertical is 21° . If the girl is holding the string 2.1 m above the ground:



- a Find x , correct to two decimal places.
- b Hence find the height h of the kite above the ground, correct to two decimal places.

